Pokhara University Faculty of Science and Technology	
Course Code: CMP 511 (3 Credit)	Full Marks: 100
Course Title: Advanced Computer Architecture	Pass Mark: 60
Nature of the Course: Theory	Total Lecture Hours: 45 hours
Level: Master	Program: ME/MSc (Computer)

1. Course Description:

- To focus on design aspects of the parallel processor and Pipelining
- To Introduce the concepts of super scalar and memory hierarchies
- To Demonstrate simulation technique

2. General Objectives:

- Understand the classes of computers, and new trends and developments in computer architecture
- Understand the concepts of pipelining, instruction set architectures, memory addressing.
- Understand the architecture and design principles of multicore processors.
- Understand the various techniques to Multithreaded architecture and multiprocessor operating system

3. Contents in Detail

Specific Objectives	Course Contents
To understand the basic concept of parallel process and its structure	1. **Introduction to Parallel Processing** (8 Hrs) 1.1. Parallel Computer Structures 1.2. Architectural Classification Parallel Processing Applications 1.3. Need of parallel programming, characteristics of parallel algorithms 1.4. parallel programming techniques 1.5. parallel programming models 1.6. parallel algorithms for multiprocessors 1.7. performance of parallel algorithms

	1.8. parallel programming languages 1.9. Problems based on parallel algorithms
Understanding the Memory and memory management system	2. **Memory technology** (7 Hrs) 2.1 Hierarchical Memory Structures, 2.2 Memory capacity planning 2.3 Cache Memories and Management 2.4 Virtual memory technology 2.5 Page replacement algorithms 2.5.1 FIFO algorithm 2.5.2 Least recently used algorithms(LRU) 2.5.3 Most frequently used algorithm (MFU)
Understand the pipelining and vector processing	3. **Principles of Pipelining and Vector Processing** (12 Hrs) 3.1 Pipelining, Instruction and Arithmetic Pipelines, 3.2 Non-linear pipeline processor 3.3 Pipeline processor classification 3.4 Arithmetic pipeline design 3.5 Instruction pipeline design 3.6 Pipeline Hazards 3.7 Dynamic instruction scheduling 3.8 Vector instruction types 3.9 Vector performance modeling 3.10 Vectorization 3.11 Vectorizing compiler design 3.12 Structures and Algorithms of Array Processors (SIMD Computers) 3.12.1 SIMD Array Processors, 3.12.2 SIMD Interconnection networks, 3.12.3 Parallel Algorithms for Array Processors Algorithm examples – matrix multiplication
Understanding the principles of Multithreaded architecture	4. **Multithreaded architecture** (7 Hrs) 4.1 Latency hiding techniques 4.2 Multithreading principles 4.3 Multithreaded architecture 4.4 Cluster computing 4.5 Neural Computing 4.6 Grid computing 4.7 Comparison of Cluster computing and Grid computing 4.8 Advantages and disadvantages of Grids
Explain the architecture and design principles of multicore processors	5. **Multicore Processor Design** (6 Hrs) 5.1. Introduction to Multicore Processors

	5.2. Design Principles of Multicore Processors: Core Integration, Instruction Parallelism, Performance Metrics 5.3. Memory Hierarchy in Multicore Systems: Cache Design (L1, L2, L3), Cache Coherence Protocols, Memory Bandwidth and Latency 5.4. Parallel Programming for Multicore Processors 5.5. Challenges in Multicore Systems 5.6. GPU architectures and general-purpose GPU (GPGPU) computing
Understanding the Multiprocessor operating system and its implementation	6. **Multiprocessor Operating System (5 Hrs)** 6.1 Classification of Multiprocessor OS 6.2 Software requirements of Multiprocessor OS 6.3 Requirement of operating system implementation 6.4 Distributed scheduler –PVM 6.5 PThread in shared memory systems

4. Methods of Instructions:
1.Lectures and conceptual explanations 2. Demonstrate the application of real-world applications 3. Assist students to gain practical skills in computer architecture 4. Individual research assignments and case study 5. Classroom discussions

5: Case Study:

An individual case study should be given to each student on computer architecture

- The CRAY family and Cray -1
- PThread in shared memory systems
- PRAM model

Internal Evaluation	**Weight**	**Marks**	**External Evaluation**	**Marks**
Attendance & Class Participation	10%	60	Semester-End examination	40
Assignments	20%			
Internal Assessment	40%			
Research/Project/Case Study Work	30%			
Total Internal		60	**Total External**	40

Full Marks: 60 + 40 = 100

Students' Responsibilities

Each student must secure at least 60% marks separately in internal assessment and practical evaluation with 80% attendance in the class in order to appear in the Semester End Examination. Failing to get such a score will be given NOT QUALIFIED (NQ) to appear for the Semester-End Examinations. Students are advised to attend all the classes, formal exam, test, etc. and complete all the assignments within the specified time period. Students are required to complete all the requirements defined for the completion of the course.

6. Prescribed Books and References

Text Books:

1. Rajiv Chopra , Advanced Computer Architecture,
2. Kai Hwang , Advanced Computer Architecture

References:

1. Hennessy Patterson, Computer Architecture, A Quantitative Approach , 5th Edition, Elsevier.
2. M.J Quinn, Designing Efficient Algorithms for Parallel Computers, McGrawHil
3. Hennessy, J. L., & Patterson, D. A. *Computer Architecture: A Quantitative Approach*.
4. Sze, V., Chen, Y.-H., Yang, T.-J., & Emer, J. S. *Efficient Processing of Deep Neural Networks*.
5. White Papers from Intel, AMD, and ARM on multicore processor technologies.